Interview

干皓丞

Interview

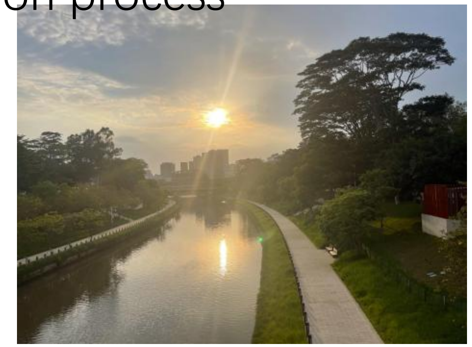
Before 2021 – Full-Stack Development and Data Analysis
Front-end and back-end development, data analysis, and Linux server maintenance.
Technologies: Angular, jQuery, JavaScript, PHP, R, CSS, HTML, and Linux Server Administration.

2021 年 之前 – 前後端開發跟數據分析
前後端開發、數據分析 : Angular、Jquery、JS、PHP、R、CSS、HTML、Linux Server 維護

Since 2021 – Artificial Intelligence and Semiconductor Applications
Generative AI, medical image segmentation, industrial image segmentation, semiconductor defect inspection, automated visual inspection, and AI applications for manufacturing and process optimization.

2021 年 之後-人工智能
生成式人工智能、醫療影像分割、工業影像分割、YOLO、UNet、MambaUNet、RF-DETR、Pytorch、Python、PyQT、C#

Models and Technologies: YOLO, U-Net, Mamba U-Net, RF-DETR, PyTorch, Python, PyQt, and C#.
My current focus is on applying artificial intelligence and computer vision to semiconductor manufacturing, equipment automation, quality inspection, and production process improvement.



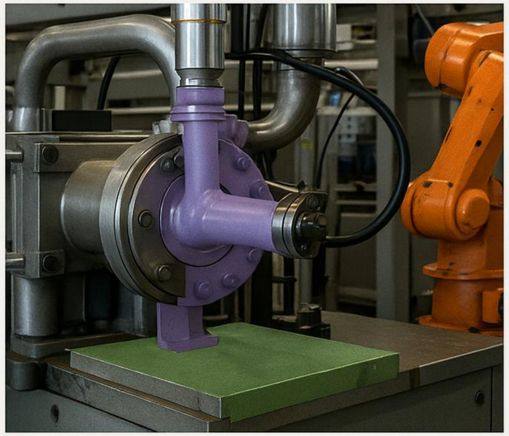
https://kancheng.github.io/

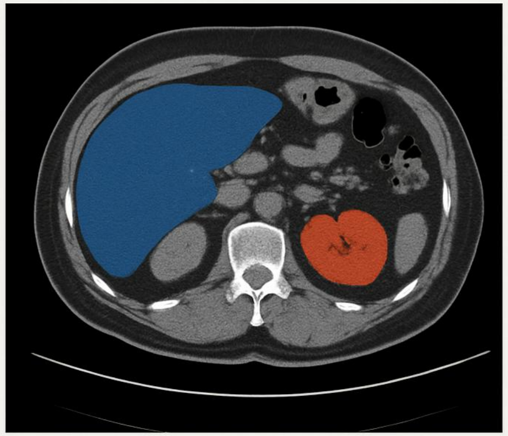
Work and Research Experience

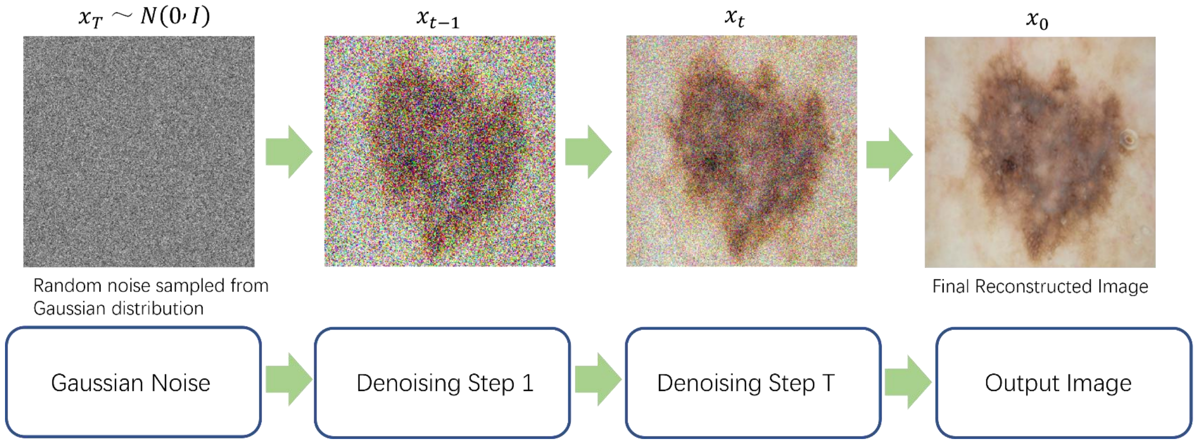
My academic research focuses on medical image segmentation.
More recently, my work has involved introducing artificial intelligence into industrial production lines, particularly for quality inspection, as well as the development, integration, and maintenance of production-line equipment.
My work covers image segmentation, object detection, synthetic data generation, and improving model generalizability across different datasets and production environments.

我在學校的研究方向是醫療影像的分割。
近期的部分是將人工智能導入工業生產線上的品質檢測。
當然範圍包含，分割、檢測、數據生成、數據泛化性

https://github.com/kancheng/happykoala







Engineering Skills

Hardware

Hardware Experience

• Linux / Ubuntu server setup
• Computer & GPU installation
• Industrial camera & lighting setup
• Network devices, RJ45 / TCP/IP
• Equipment connection & verification

Troubleshooting Process

• Check power and network first
• Verify camera and lighting
• Confirm image quality
• Test software and hardware settings
• Find root cause step by step

Manufacturing Proof

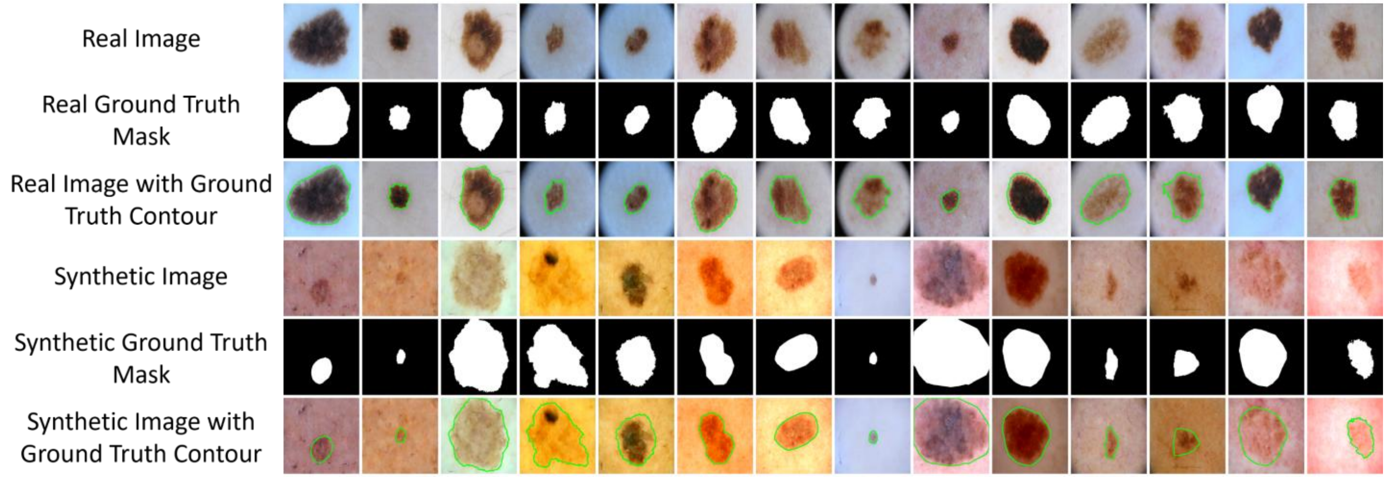
• Industrial image inspection systems
• Production-line workflow simulation
• Quality inspection process
• ONNX deployment
• System integration and testing

Software + Hardware + Manufacturing

Work and Research Experience

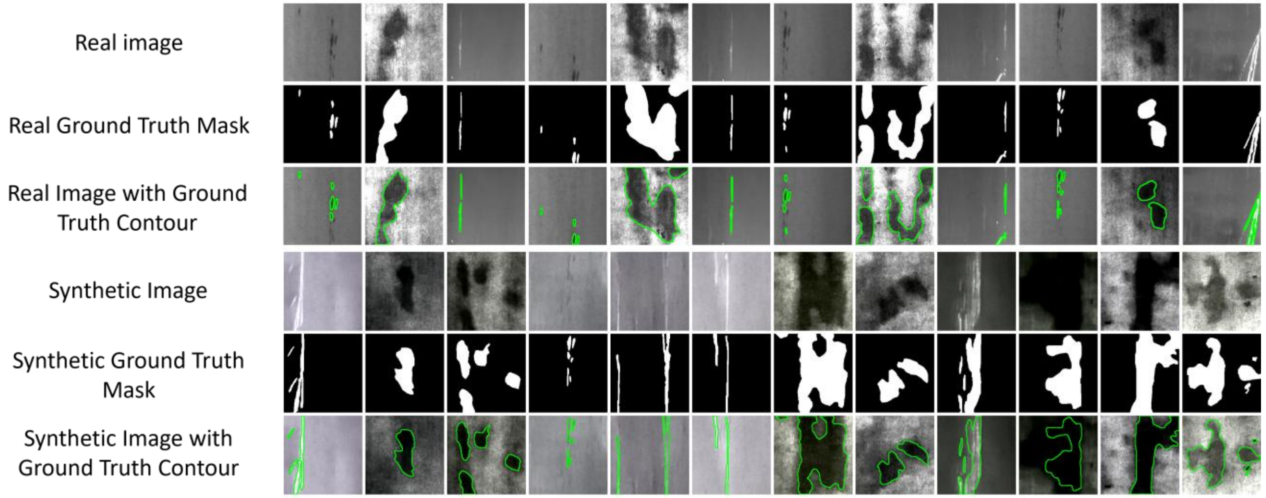
Medical

醫療



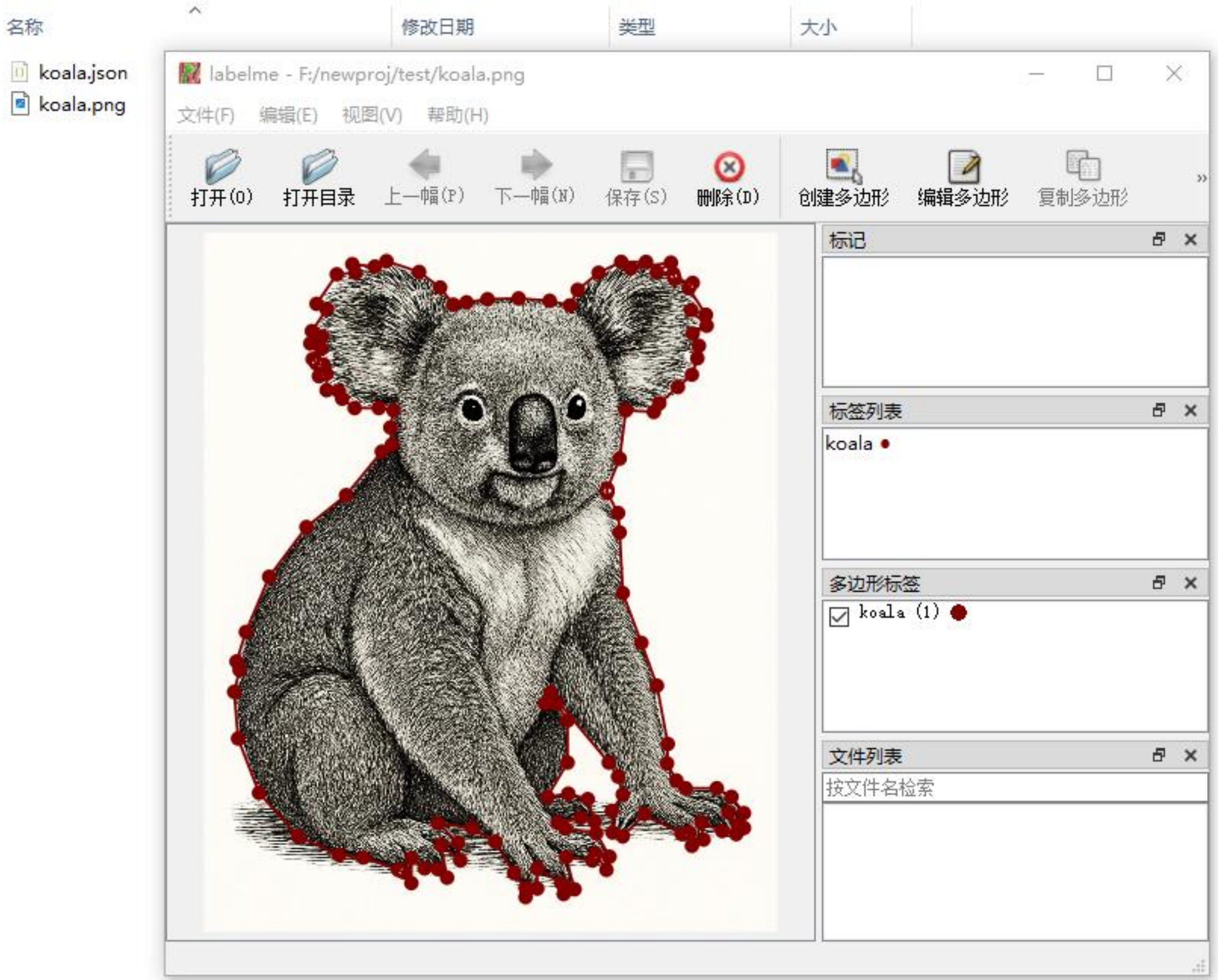
Industrial

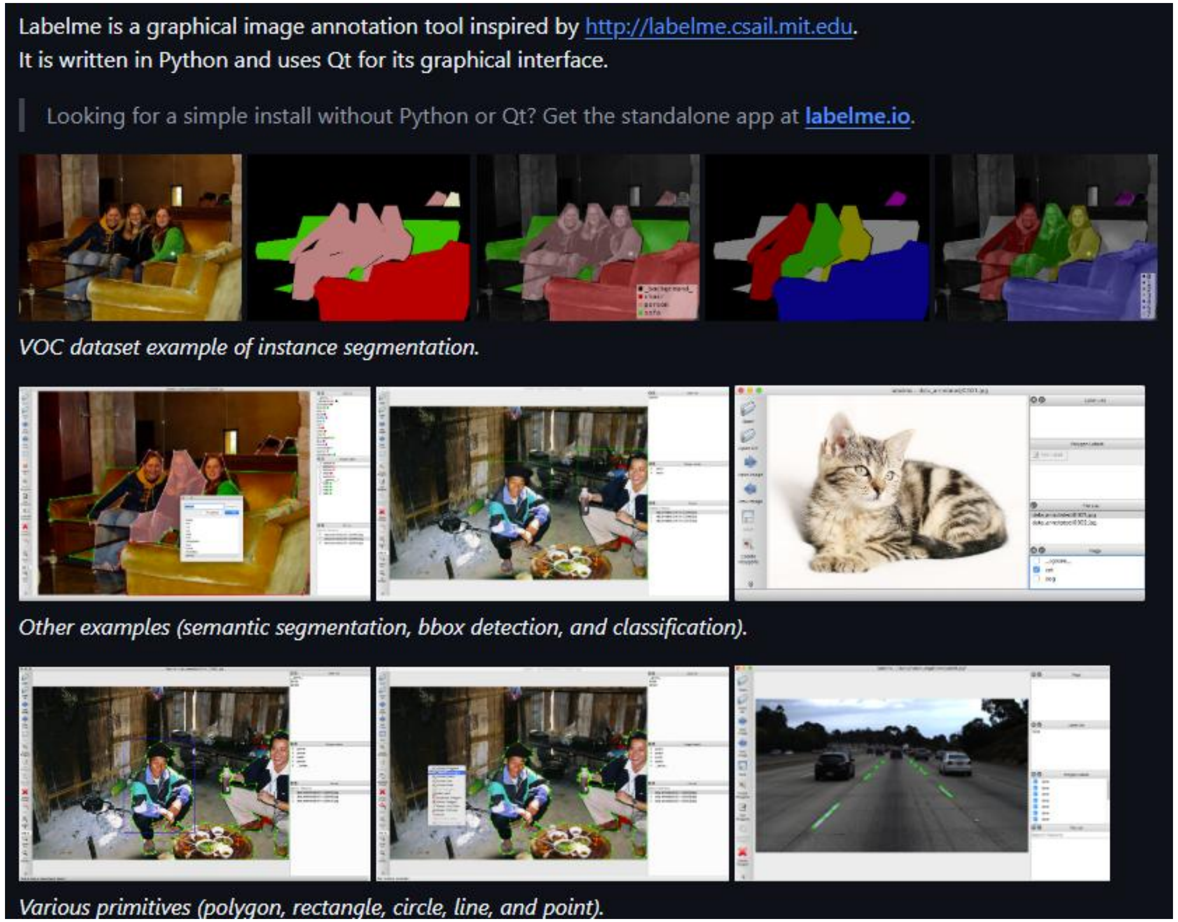
工業



開源貢獻

https://github.com/wkentaro/labelme
https://github.com/kancheng/labelme



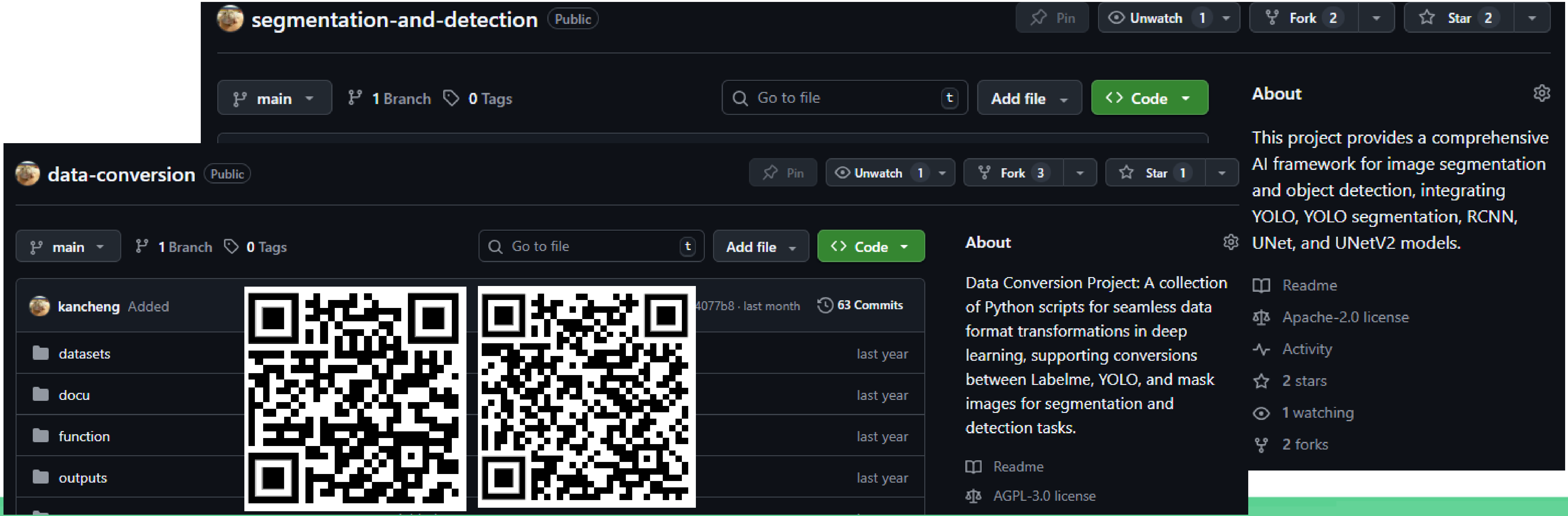


數據標註跟基本分割

https://github.com/kancheng/coco128-seg-labelme-format

數據分割的部分，可自己訓練模型。EX：Labelme 、YOLO 、Masks 格式
在使用 Labelme 工具標註完後，轉換成 YOLO、Masks 格式。然後切記要在訓練中加入OK 圖。

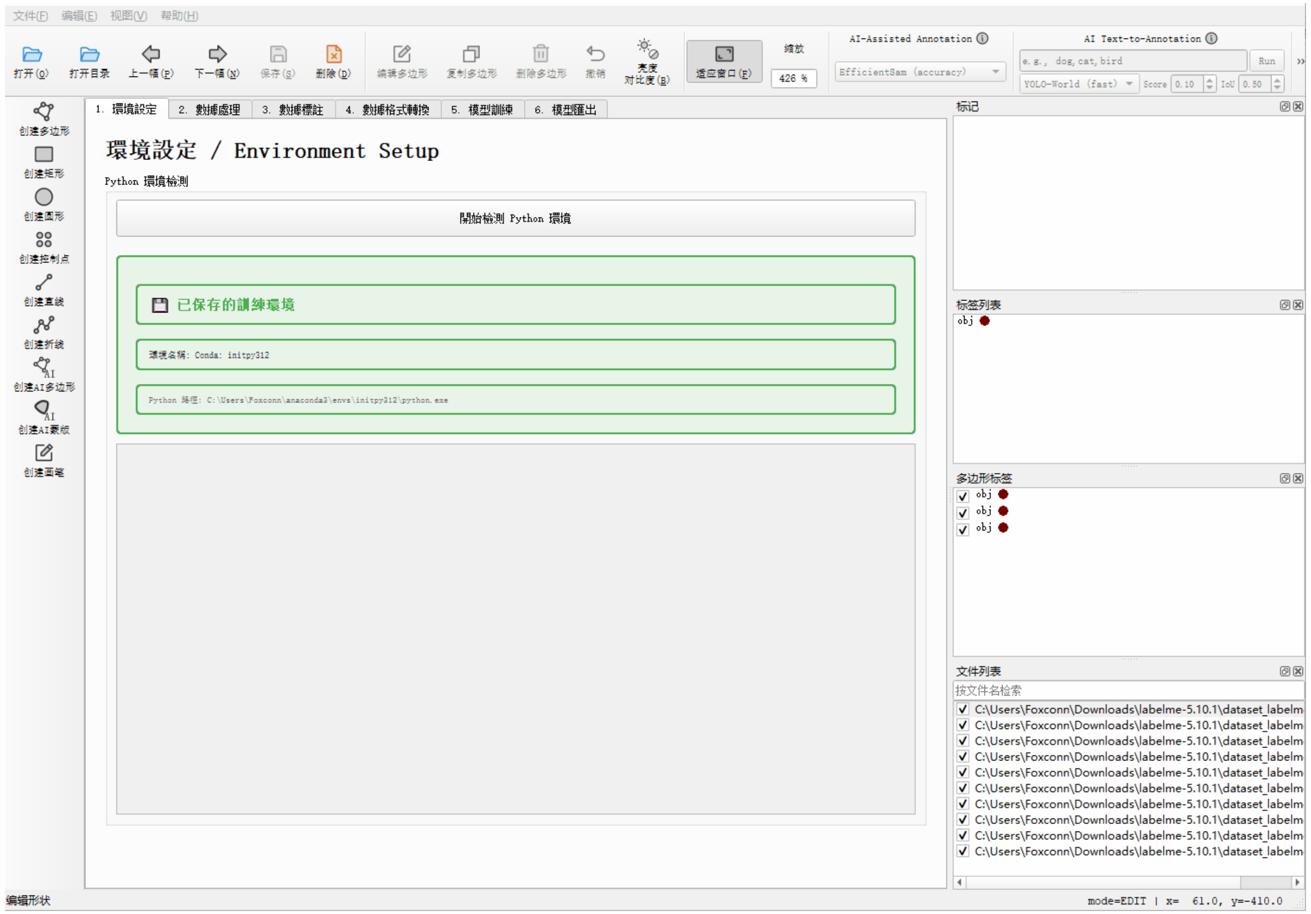
https://github.com/kancheng/data-conversion
https://github.com/kancheng/segmentation-and-detection



PyQT - Capybara

https://github.com/kancheng/labelme/tree/dev/capybara

根據 Labelme 進行為基礎，徹底改寫為 GUI 操作的影像分割與檢測訓練工具。
其功能包含有數據轉換、增強、訓練、匯出為 onnx.



C# - YoloIndustrySegmentation

YOLO 模型訓練為 onnx，並撰寫一個基於 C# 的 GUI 檢測方案。

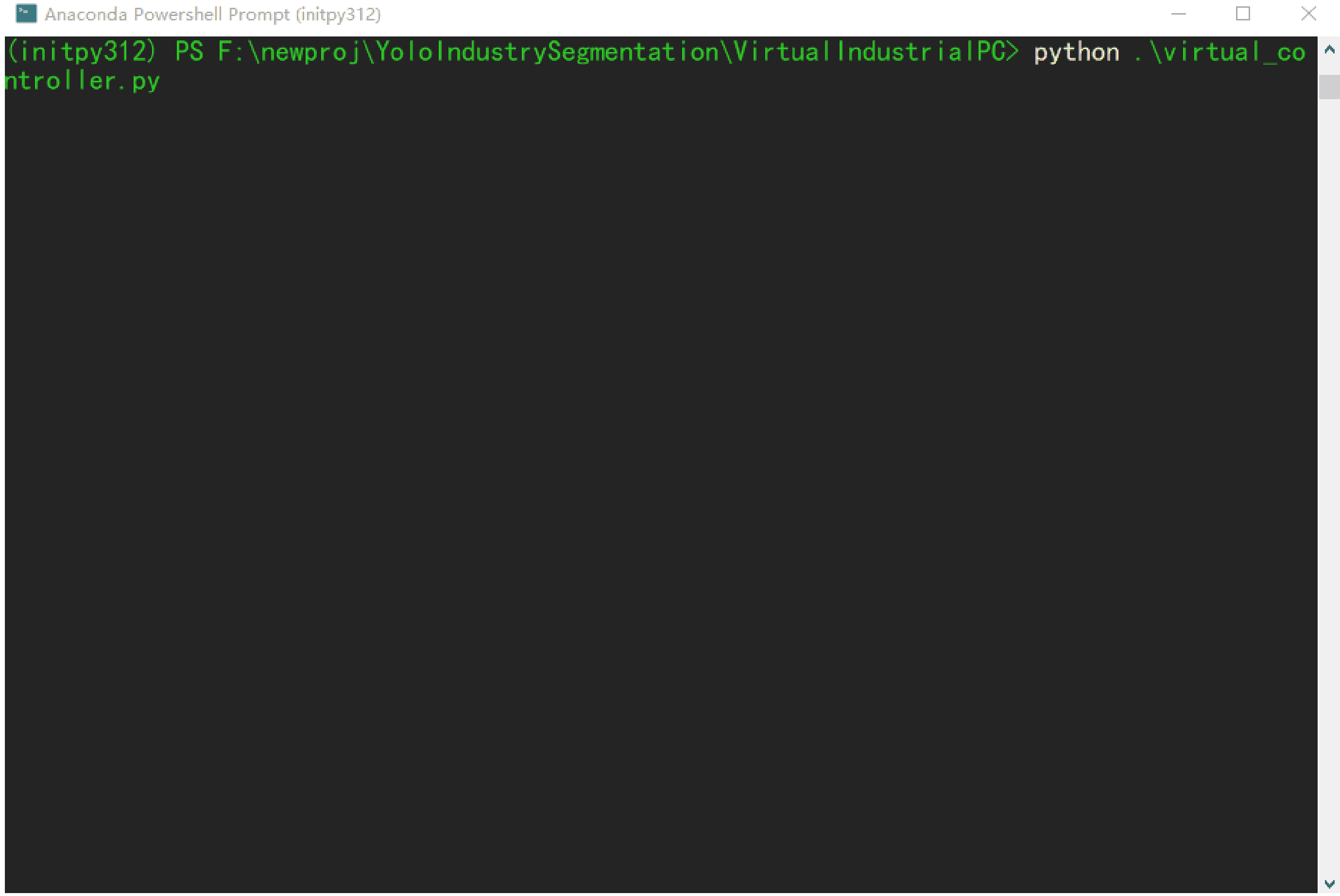
https://github.com/kancheng/YoloIndustrySegmentation

C#

Form1
基本运动
速度设置
距离设置
X正
X反
X停止
速度
10000
距离
10000
目标位置
状态
Y正
Y反
Y停止
速度
10000
距离
10000
目标位置
状态
Z正
Z反
Z停止
速度
10000
距离
10000
目标位置
状态
X轴回原点
Y轴回原点
Z轴回原点
初始化系统
关闭系统
回原点
等待初始化.
停止
急停
正向联动
负向联动
直线插补
用户：
逻辑位置：
实际位置：

Python - YoloIndustrySegmentation

Production Line Simulation- Python 該項目有一個模擬生產線行為的 Python。

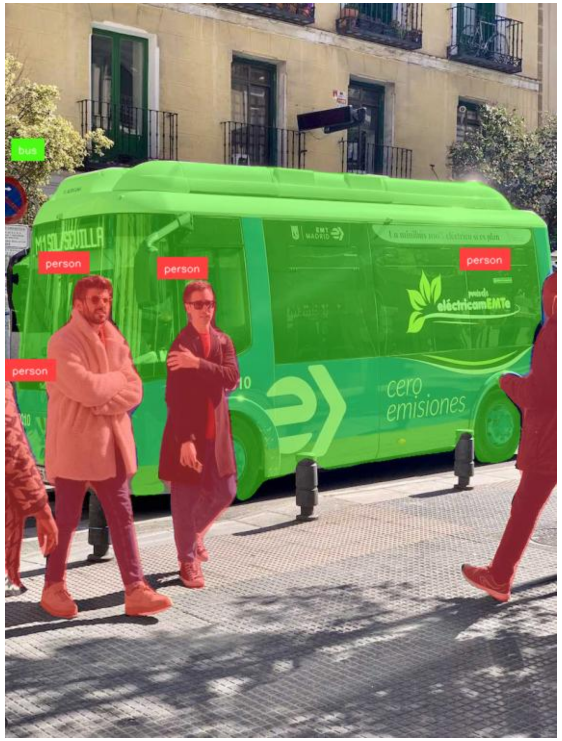


https://github.com/kancheng/YoloIndustrySegmentation

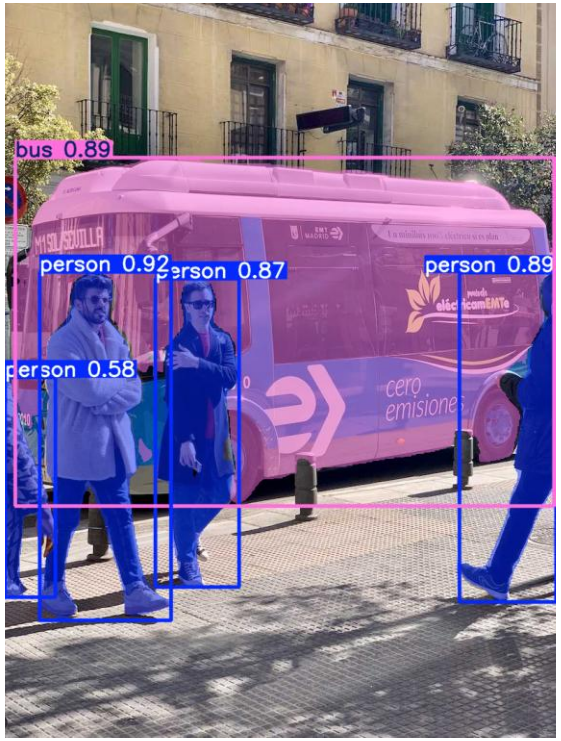
CV–YOLO & RF-DETR

RAW

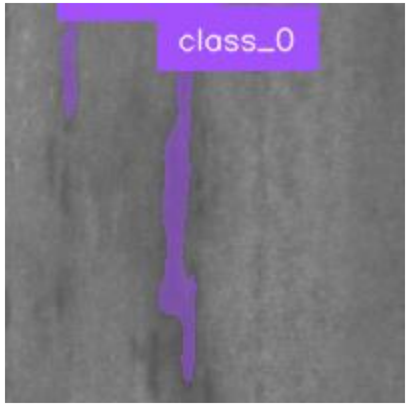
RF-DETR



YOLO



RF-DETR 金屬分割



https://github.com/kancheng/rfdetr-seg-train-predict

https://github.com/kancheng/yolo-vs-detr-demo

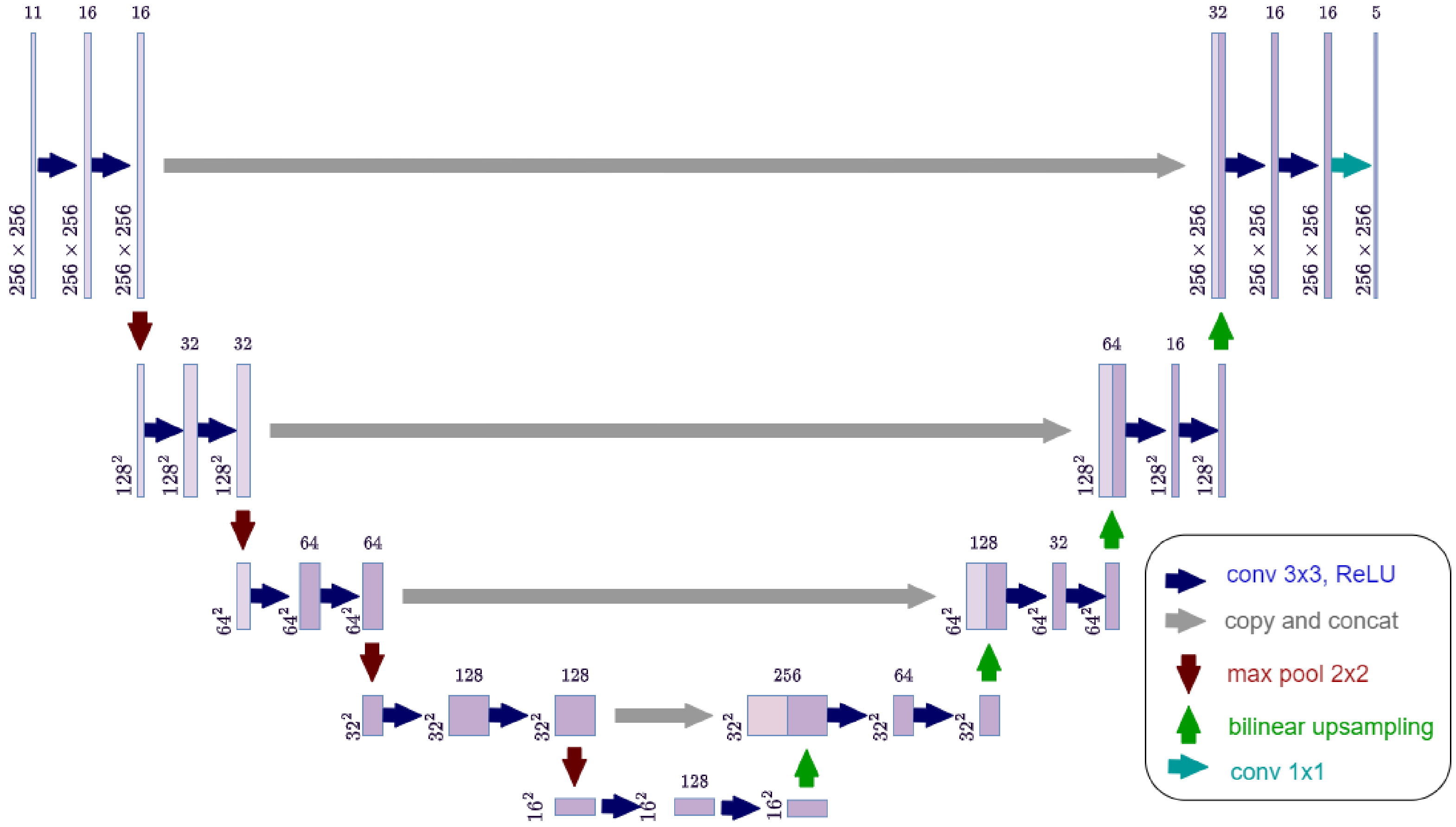
分割模型

UNet, U2Net,TranUnet, VMUnet, ResUnet,UNet++,Unet+++

最早於 2015 年由 Olaf Ronneberger 等人提出，其論文名稱為《U-Net: Convolutional Networks for Biomedical Image Segmentation》。

結構呈現對稱的 U 型，主要由兩個部分組成：

1. 編碼器（Encoder / Contracting Path）
2. 解碼器（Decoder / Expansive Path）



初步消融實驗

UNet, U2Net,TranUnet, VMUnet, ResUnet,UNet++,Unet+++

Type	mIOU	DSC	Accuracy
UNET	0.6877	0.8150	0.9190
VMUNET	0.7944	0.8854	0.9443
U2NET	0.7296	0.8437	0.9232
UNET++	0.5968	0.7475	0.8682
UNET+++	0.7177	0.8356	0.9246
VMUNETV2	0.7812	0.8771	0.9403
HVMUNET	0.7990	0.8883	0.9443
TRANSUNET	0.6277	0.7713	0.8982
RESUNET	0.6061	0.7547	0.8861
RESUNET++	0.6677	0.8007	0.9090

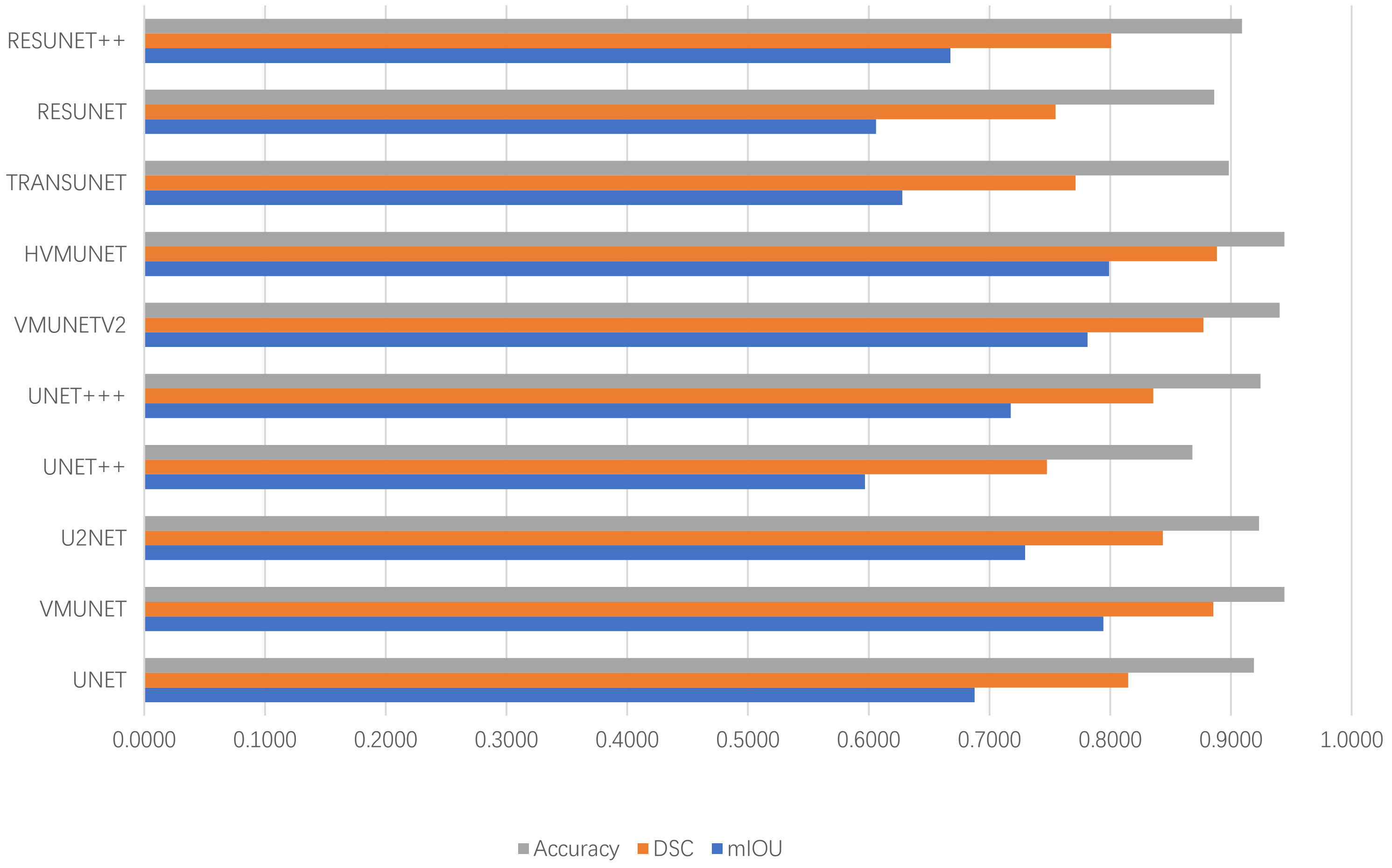
基礎模型，表現穩定但非最佳。

優秀表現，三項指標皆在前列。

所有指標皆為最佳或並列最佳，表現最強。

初步消融實驗

UNet, U2Net,TranUnet, VMUnet, ResUnet,UNet++,Unet+++



mIOU（平均交併比）：評估預測區域與真實區域的重疊程度。數值越高表示模型預測越準確。

DSC（Dice 系數）：衡量兩區域相似性的指標，常用於醫療影像分割。越接近 1 越好。

Accuracy（準確率）：預測正確像素所佔總像素的比例。對於背景主導的資料集，準確率容易偏高，但不一定能真實反映分割品質。

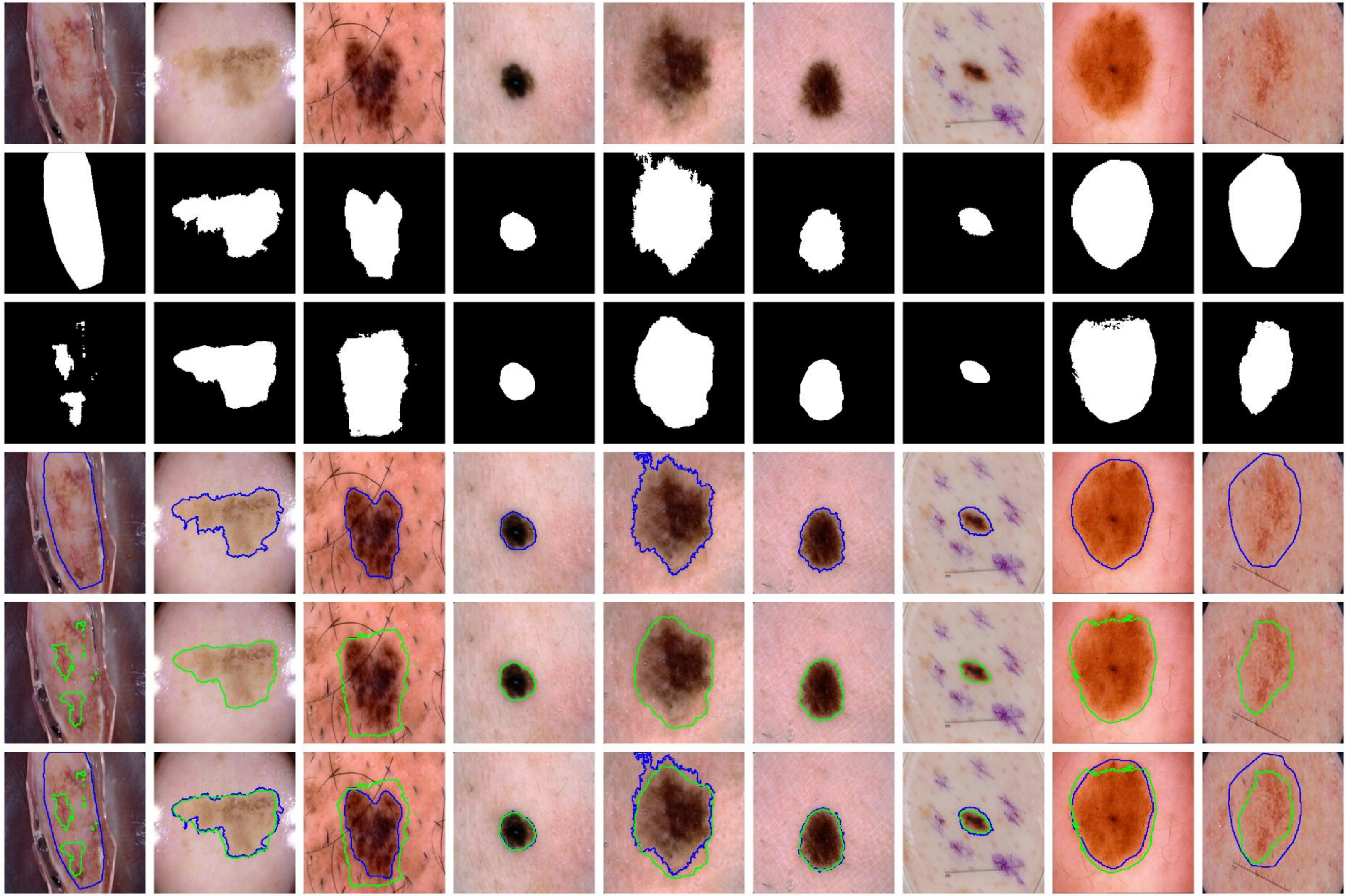
實驗呈現

UNET

原始影像
Image

標註真值
ground truth

模型預測結果
prediction



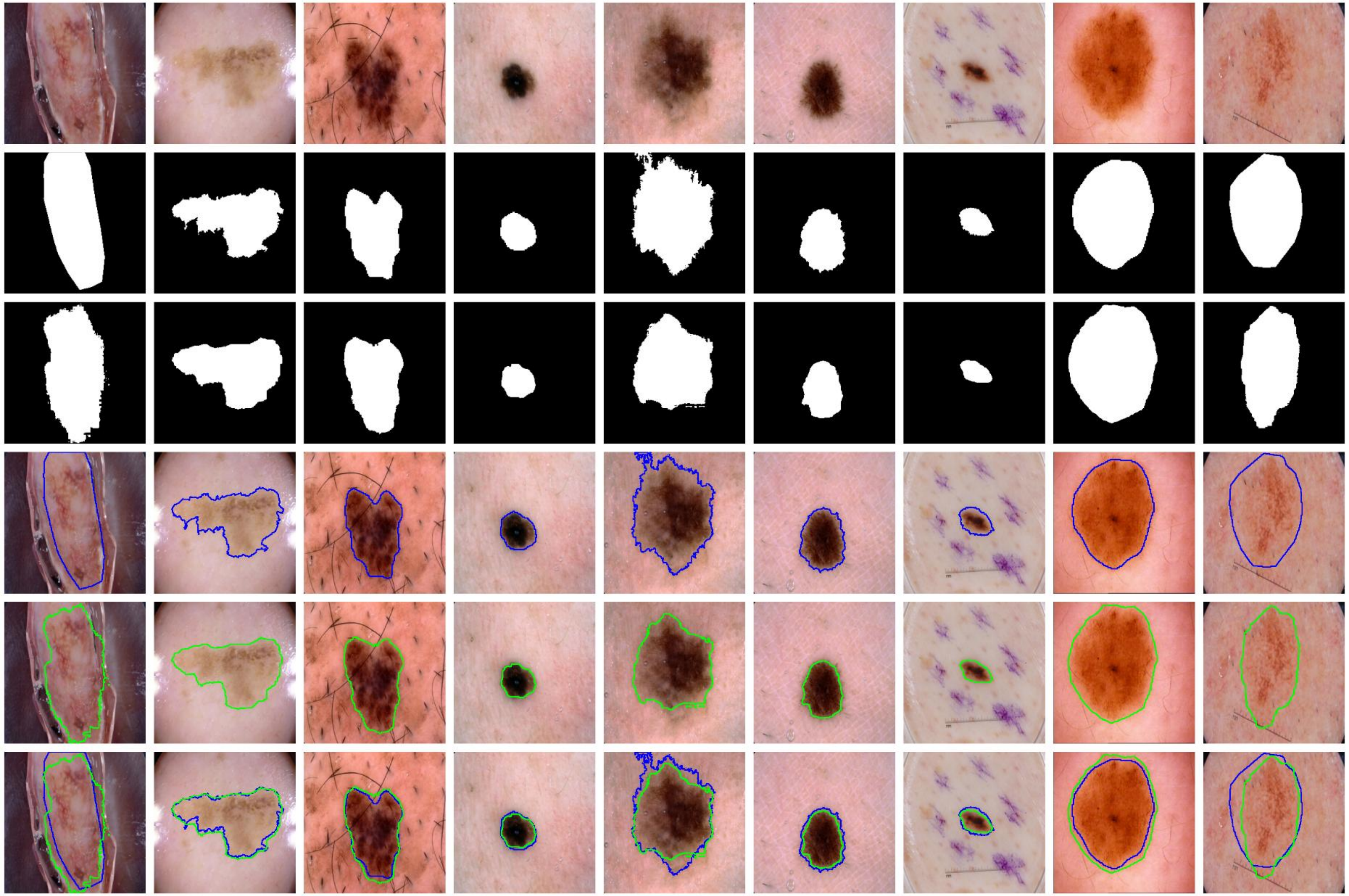
實驗呈現

VMUNET

原始影像
Image

標註真值
ground truth

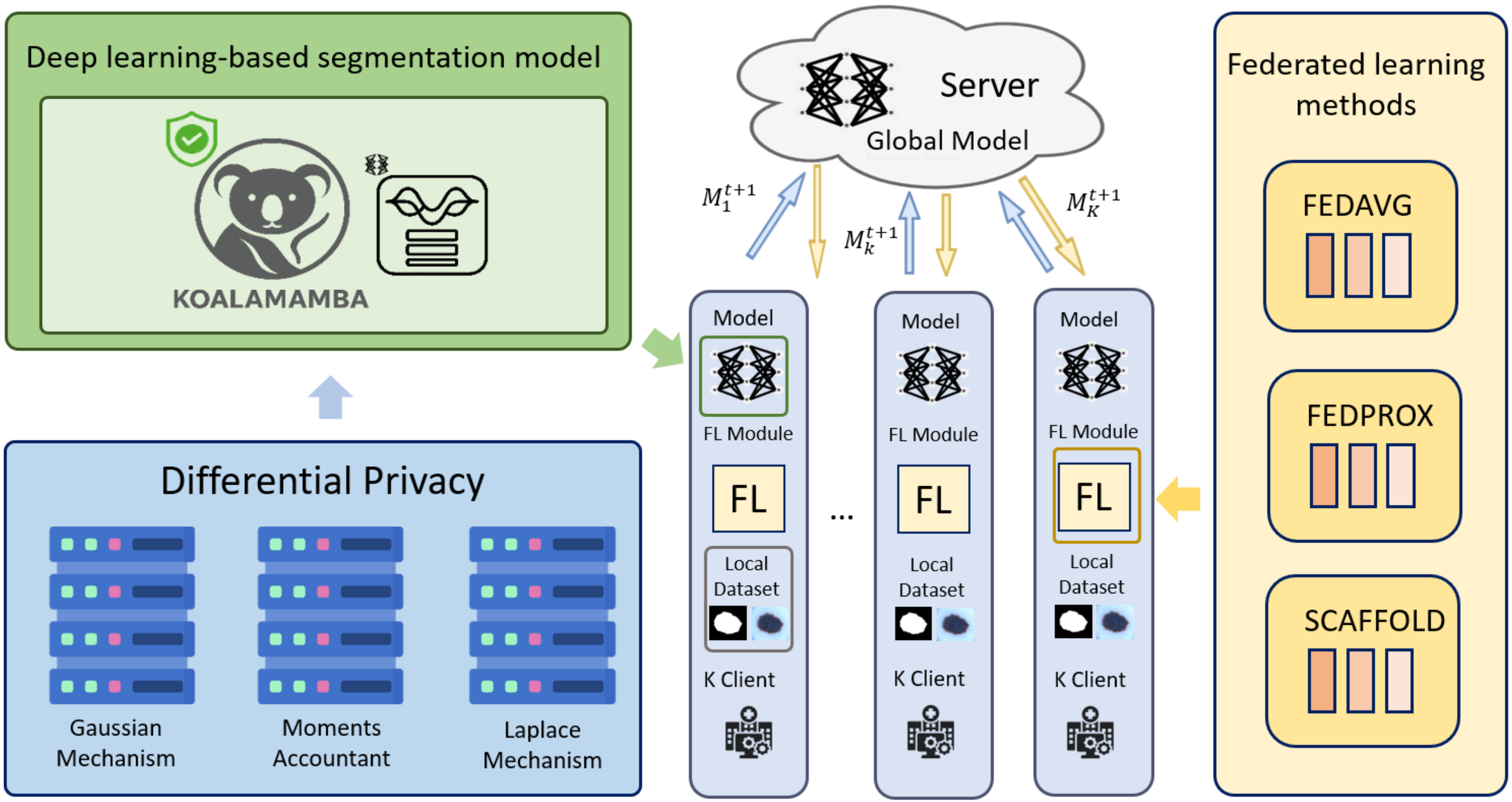
模型預測結果
prediction



發表

1. **Haocheng Kan**, Yuesheng Zhu, Guibo Luo and Hanwen Zhang. OTTER: Optimized Training with Trustworthy Enhanced Replication via Diffusion and Federated VMUNet for Privacy-Aware Medical Segmentation. [C]//The 2025 International Conference on Information and Communications Security (ICICS). 2025. **(CCF-C)**

2. **Haocheng Kan**, Yuesheng Zhu, Guibo Luo and Mingpei Cao. KoalaMamba: A Federated Method with Vision Mamba UNet for Privacy-Preserving Medical Segmentation. [C]//WCCI 2026. (**CCF-C**)



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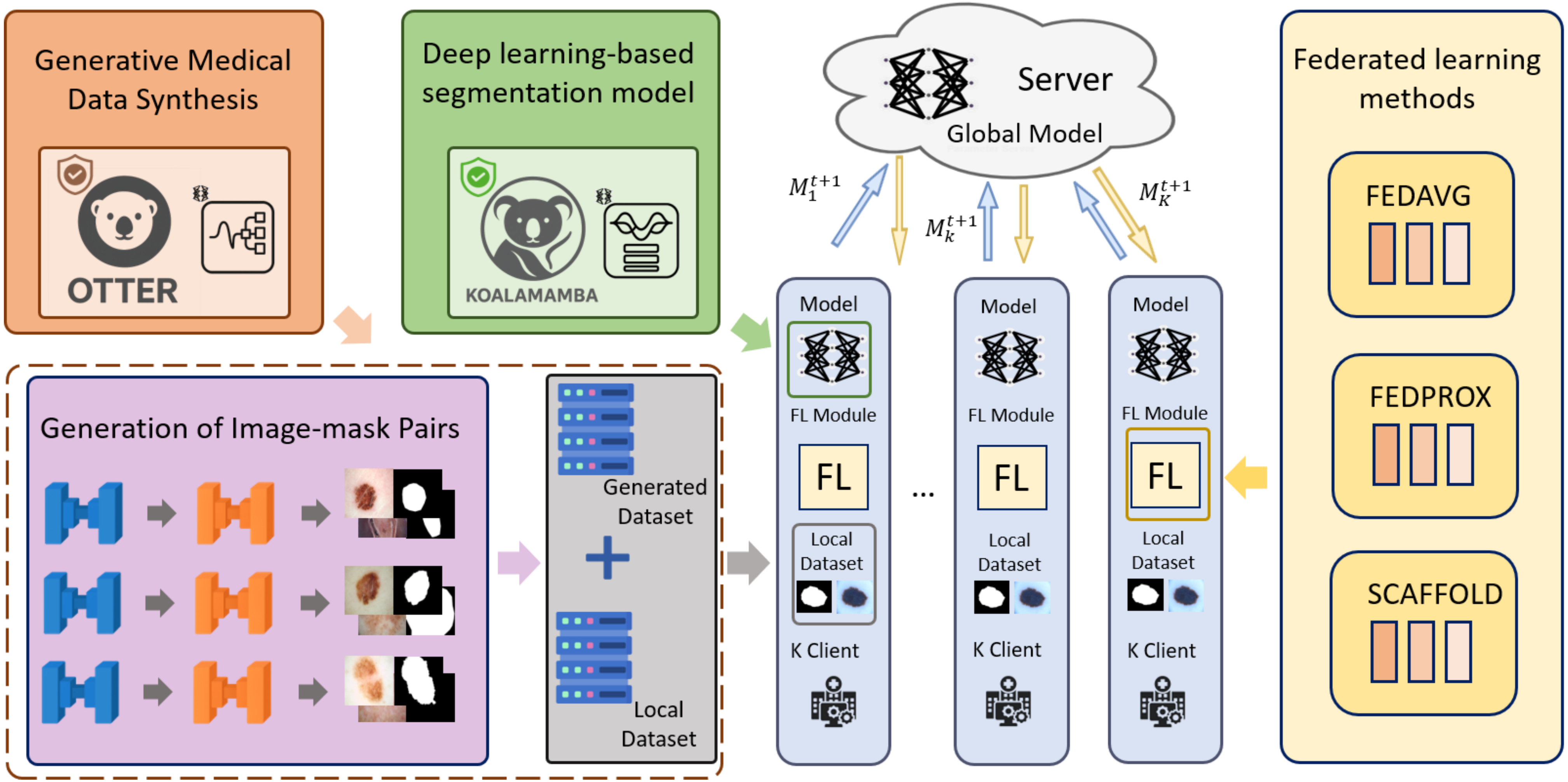

結論發表

醫療

1. **Haocheng Kan**, Yuesheng Zhu, Guibo Luo and Hanwen Zhang. OTTER: Optimized Training with Trustworthy Enhanced Replication via Diffusion and Federated VMUNet for Privacy-Aware Medical Segmentation. [C]//The 2025 International Conference on Information and Communications Security (ICICS). 2025. **(CCF-C)**
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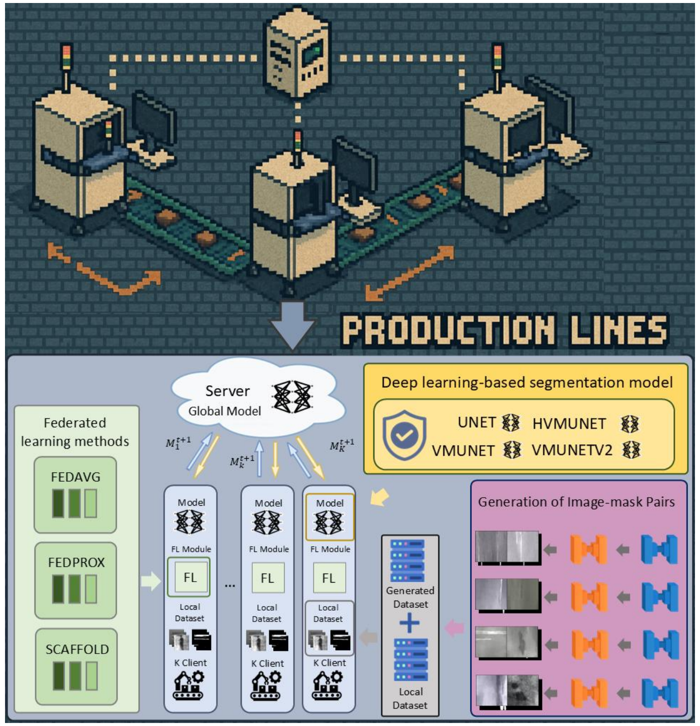
工業

投稿中 ...

RABBIT: A Privacy-Aware Diffusion and VMUNet-Based Segmentation Framework for Multi-Factory Industrial Defects



End

https://github.com/facebookresearch/dinov3

Linux 底層開發、嵌入式設備、ONNX、DINOV3